



DATA SCIENCE SEMINAR PROJECT-2 (WINTER) 2025

**CROSS DATASET COVID-19 CLASSIFICATION: A
COMPARATIVE STUDY OF PATIENT RISK AND SEVERITY
PREDICTION**

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GROUP I

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INTRODUCTION

As of July 2019, the COVID-19 virus had infected approximately 16 million people and killed about 660,000 people (Dong et al., 2020). The virus had also spread widely to more than 27 countries and to all the provinces of China by February 17, 2020. Symptoms of the deadly disease include coughing, fever and fulminant pneumonia (Singhal, 2020).

Several studies have also been done to study the disease progression of the covid19 disease (Syeda et al., 2021). His analysis showed that age and minimum oxygen encounters were the signifying predictors of disease progression identified in models.

Covid-19 pneumonia was also a significant risk to high mortality rates during covid as it differs from general pneumonia.

LITERATURE REVIEW

There have been several similar studies done on the use of machine learning in predicting covid outcomes. One of such studies by Booth et al. (2021) was aimed at developing a prognostic model for COVID-19 mortality using serum biomarkers and machine learning. The study focused on identifying high-risk patients up to 48 hours before death. The study used Support Vector Machine with Shapley additive explanations in order to enhance interpretability of the model which is paramount in the medical field. The SVM model achieved 91% sensitivity, 91% specificity, and an AUC of 0.93. The Strengths of the paper include: it had high predictive performance with a small set of biomarkers, usage of SHAP values enhanced model interpretability, exclusion of data within 48 hours of death improved clinical utility. Some of the

gaps in the paper are that it was a single-center study, potentially limiting generalizability and had a relatively small sample size, especially for deceased patients (n=43) when the covid 19 pandemic killed so many and it did not account for dynamic changes in biomarkers over time.

Another study by Moulaei et al. (2022) compared multiple machine learning algorithms in order to select the best-performing model for the prediction of COVID-19 mortality using data from hospital admissions. The study used models like XGBoost, Random Forest, MLP and logistic regression etc. The study also used (Synthetic Monority Over Sampling Techniques (SMOTE) to address the imbalance in the dataset. In this study, the Random Forest performed better than the other datasets with a 95.03% accuracy, 90.70% sensitivity and a AUC-ROC curve value of 99.02%. Some of the strengths of the paper included that it had large sample size (n=1,500) compared to previous studies as well conducting a comprehensive feature selection using Delphi surveys and ML techniques. It also used SMOTE to improve model performance on imbalanced data. Some of the gaps include that it used a retrospective design with potential documentation biases and that it had Limited generalizability due to single-center data. Lastly, it did not incorporate imaging or longitudinal lab trends.

Another study by Heidari et al. (2022) evaluated techniques used in the managemnet of covid-19 such as forecasting case trajectories using Long Short-Term Memory and Random Forest as well as predicting mortality risk as well as other things. The strengths of this paper include that it has comprehensive coverage as the review spans diverse applications, from medical imaging to socio-economic analysis. It also had methodological rigor as it used a systematic literature review

(SLR) framework to analyze 37 studies with very clear clarity on the specifics of the inclusion criteria as well as technological intergration by using IoT, cloud computing. The gaps in the paper include that the datasets used for study has large variability in their sizes which means that there could be issues in the ability of the results of this paper to be generalized for other uses. The study also used retrospective design and lacked prospective testing and thus had no basis for real world validation.

Lastly, Dutta and Bandyopadhyay (2020) used an LSTM-GRU-RNN model to verify COVID-19 case classifications (positive, negative, deceased, released). The accuracy of the model could be summarized as follows: 87% for confirmed cases, 67.8% for negative cases, 62% for deceased, and 40.5% for released cases. The rmse in the study results ranged from 4.16% (deceased) to 49.4% (negative cases). Also a key result was that the combined LSTM-GRU model performed far better than the standalone LSTM or GRU models, showing the superior handling of sequential data. The strengths of this paper include that it used novel architecture as Hybrid LSTM-GRU improved performance over the initial one model approaches. The study also showed ability to be used in real world setting as it can serve as an automated tool to cross-validate diagnoses, reducing workload for physicians. Some of the gaps in the study include limited scope of data as the study only concentrated on south korea and the time was only between january and march 2020. The model also had poor results for the accuracy (40.5%) of released cases suggesting that it cannot accurately predict when a patient will recover. Lastly, the dataset lacked real time updates as such cannot be widely applied to the ever-changing realities of pandemic evolution.

METHODOLOGY

Firstly, Explanatory Data Analysis was done on the dataset and we made key discoveries that would form the basis of our modelling process:

Dataset 1: Symptom and Severity Data (316,800 records)

- **Key Features:** The dataset contained detailed symptom information (fever, cough, breathing difficulty) and severity classifications
- **Unique Aspects:** The unique aspect of the data included demographic data (gender) and contact history of those being studied.
- **Size:** There were about 27 columns including multiple symptom indicators and severity levels.

Dataset 2: India-Specific Case Data (18,110 records)

- **Key Features:** This dataset tracked COVID-19 cases, deaths, and recoveries across Indian states
- **Unique Aspects:** It also included time-stamped records and foreign national cases.
- **Size:** It had 9 columns with temporal and geographical dimensions

Dataset 3: Global WHO Data (49,068 records)

- **Key Features:** It had extensive worldwide case data organized by WHO regions
- **Unique Aspects:** It contained key geographical coordinates (lat/long) and regional breakdowns which were extremely helpful in later analysis.
- **Size:** It had 10 columns with international scope.

Dataset 4: Vaccination Data (86,512 records)

- **Key Features:** It tracked vaccination progress across countries
- **Unique Aspects:** It ncluded per-capita metrics and daily vaccination trends
- **Size:** It had 15 columns focusing on vaccination metrics

Dataset 5: Patient Clinical Data (1,048,575 records)

- **Key Features:** it gave detailed patient-level clinical information

- **Unique Aspects:** It included comorbidities, ICU status, and treatment details
- **Size:** It had 21 columns with comprehensive medical indicators

Next, we calculated the percentage of missing values for each column. Dataset 4 showed significant missingness in vaccination metrics (49-59% for some columns). Dataset 3 had 70% missing values in Province/State field while other datasets were remarkably complete (<1% missing values). For descriptive statistics, we created a comprehensive statistical summary for all numeric columns. This included measures of central tendency and dispersion (mean, std, min/max, quartiles).

- A) For Dataset 1: The most common symptoms showed dry cough was the most common with 56.25%, fever in 31.25%.
- B) For Dataset 5: The average patient age was 41.8 years with wide range (0-121 years).

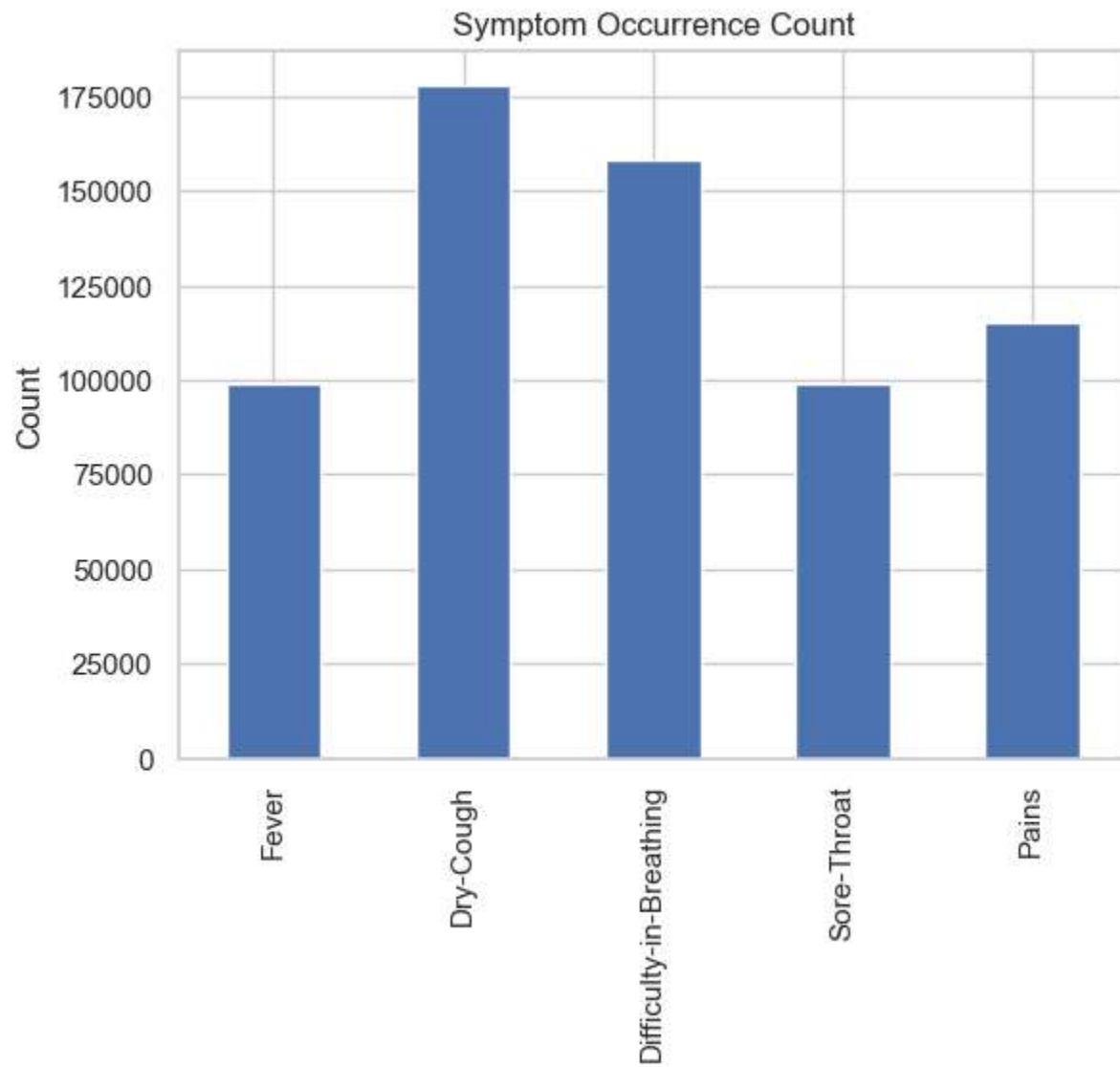
Next we did analysis on the categorical values by, checking the frequency distributions for categorical variables:

- A) Dataset 1: There was a balanced representation across countries and gender categories.
- B) Dataset 2: It showed that most records were timestamped at 8:00 AM (16,472 of 18,110)
- C) Dataset 3: Here Europe had the most records (15,040), South-East Asia fewest (1,880).

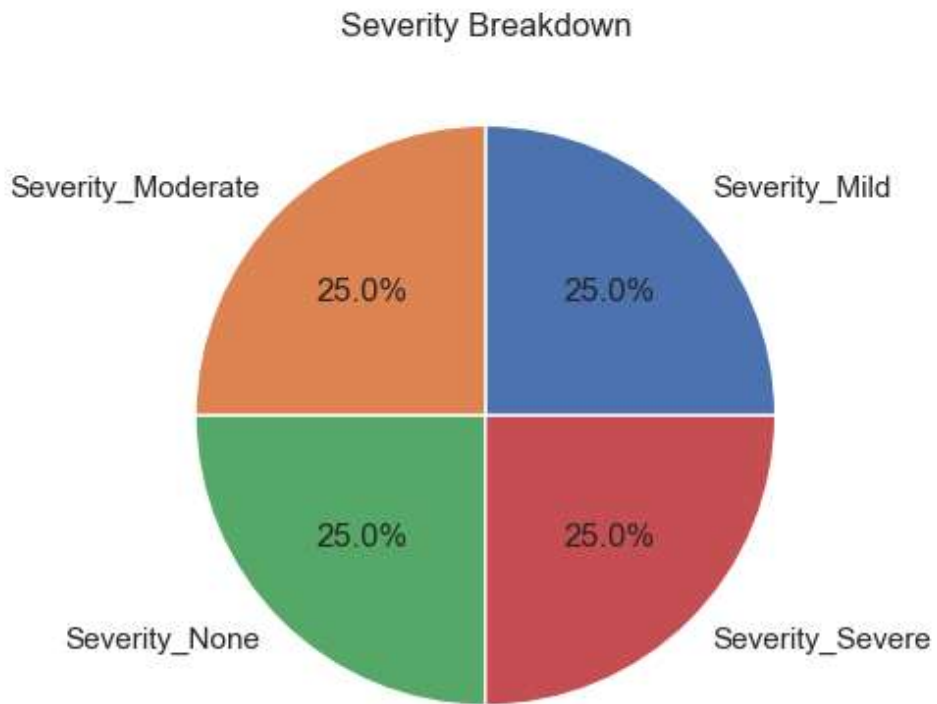
Then we did a comprehensive visualization for each dataset:

Dataset 1:

Symptom Occurrence: Bar chart showing dry cough most prevalent

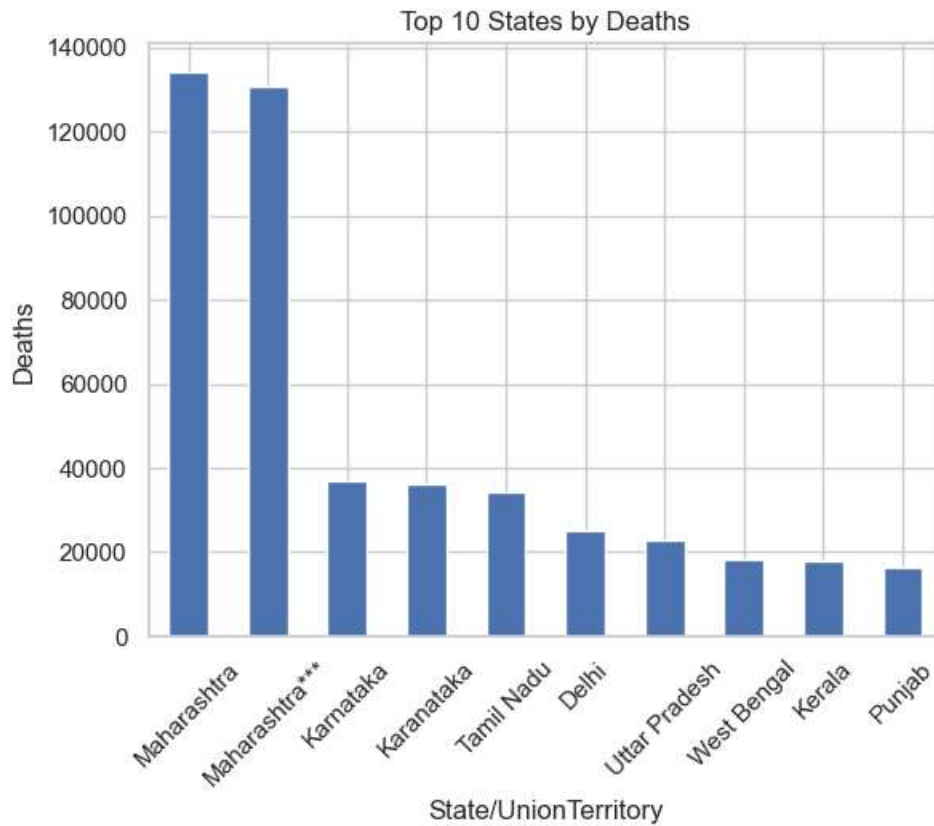


Severity Distribution: Pie chart revealing balanced severity classes (25% each)

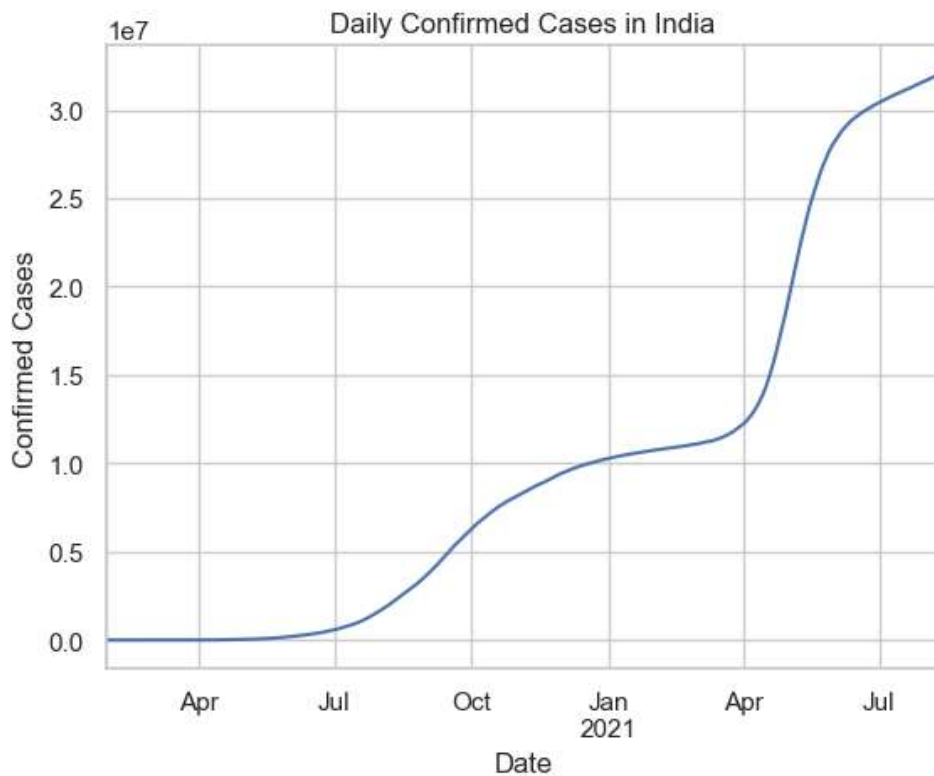


Dataset 2:

- **Geographical Impact:** Top 10 Indian states by death count

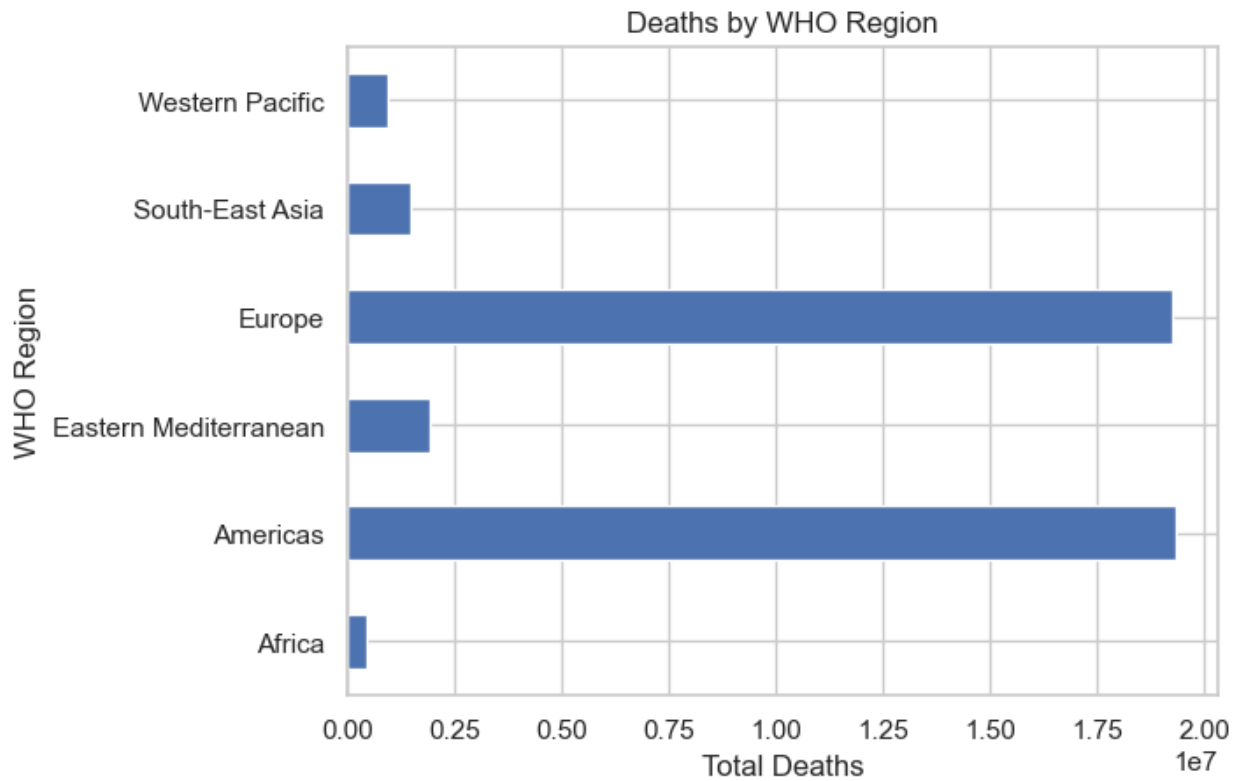


- **Temporal Trend:** Daily confirmed cases progression in India

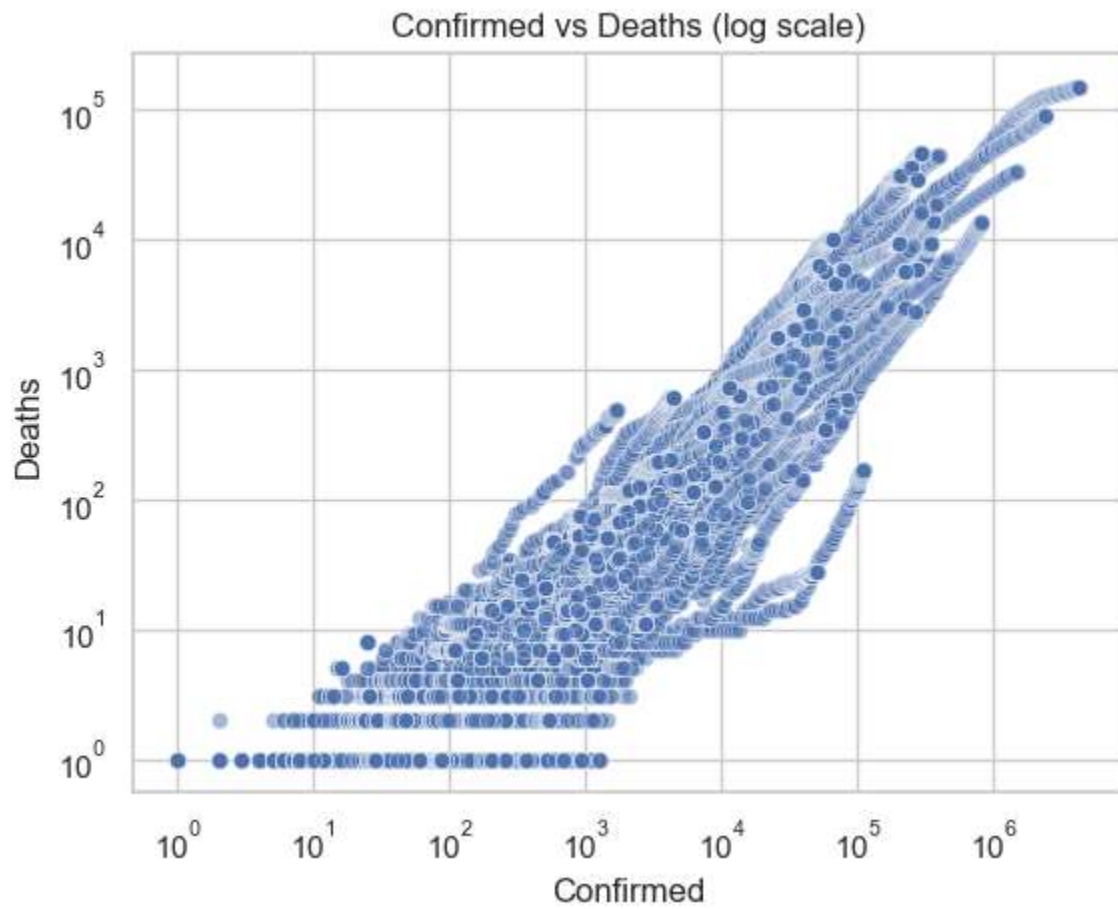


Dataset 3:

- **Regional Analysis:** Horizontal bar chart of deaths by WHO region

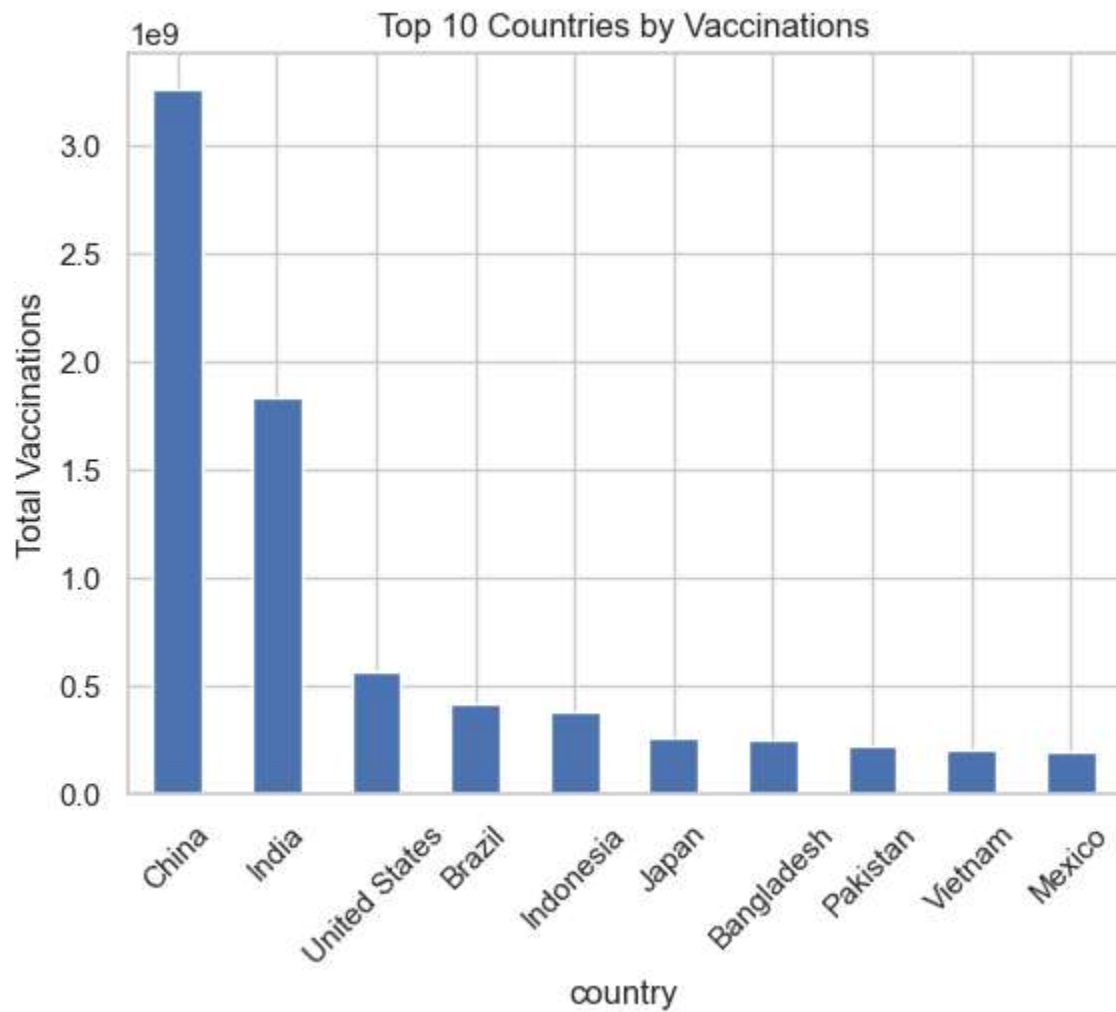


- **Case-Fatality Relationship:** Log-scale scatterplot of confirmed cases vs deaths

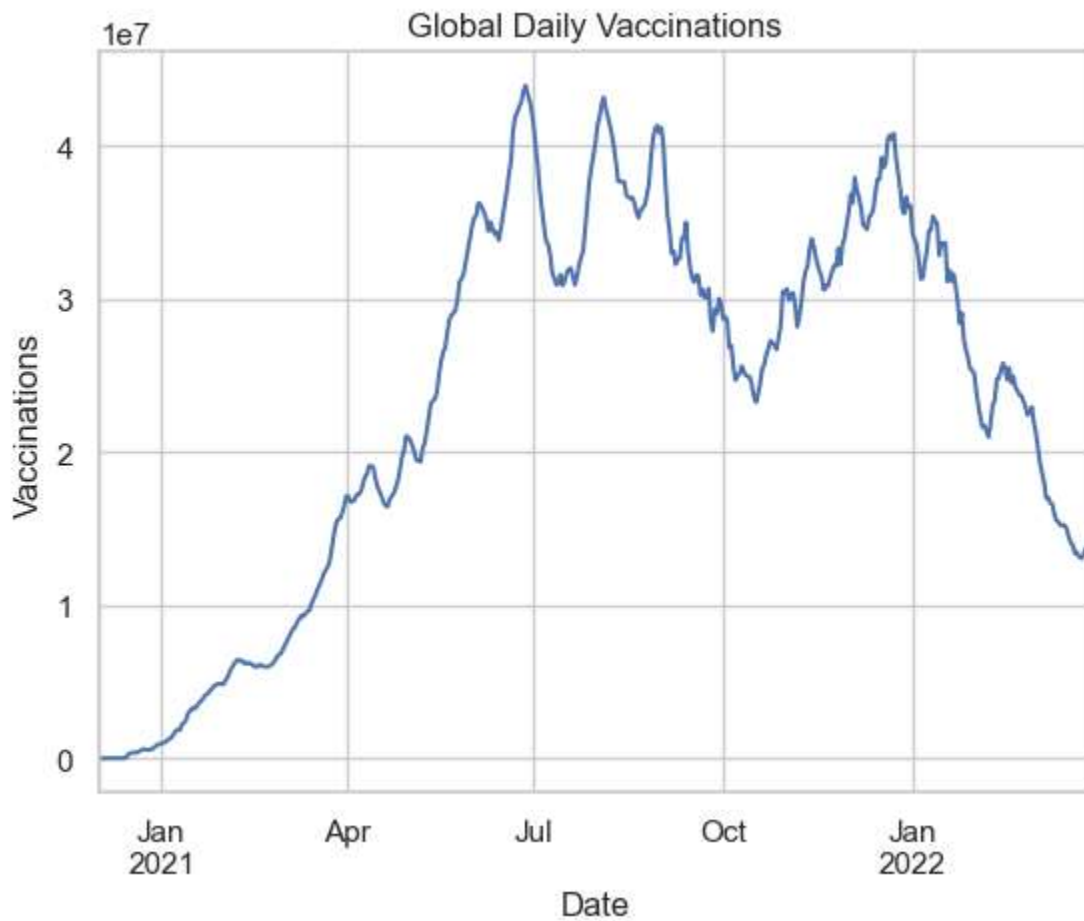


Dataset 4:

- **Vaccination Progress:** Top 10 countries by total vaccinations

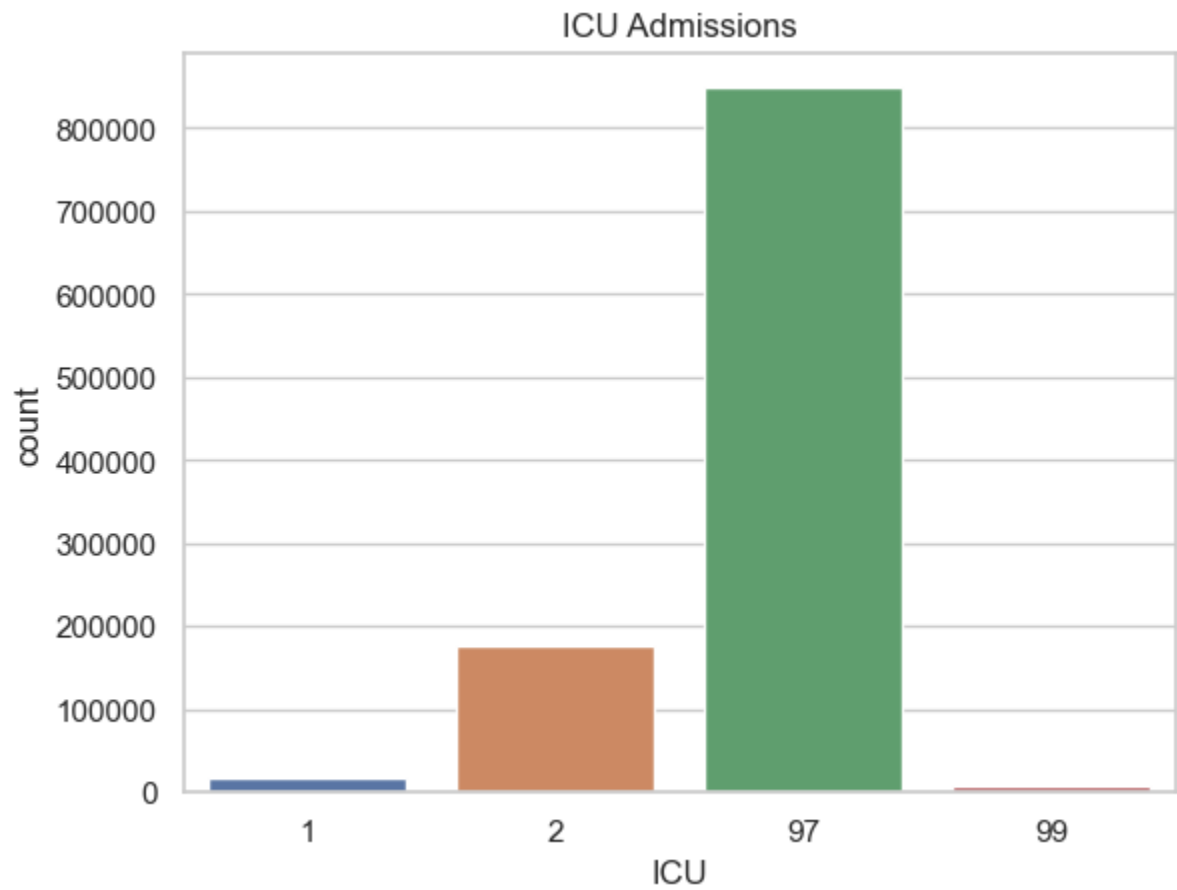


- **Global Trends:** Time series of daily vaccination rates

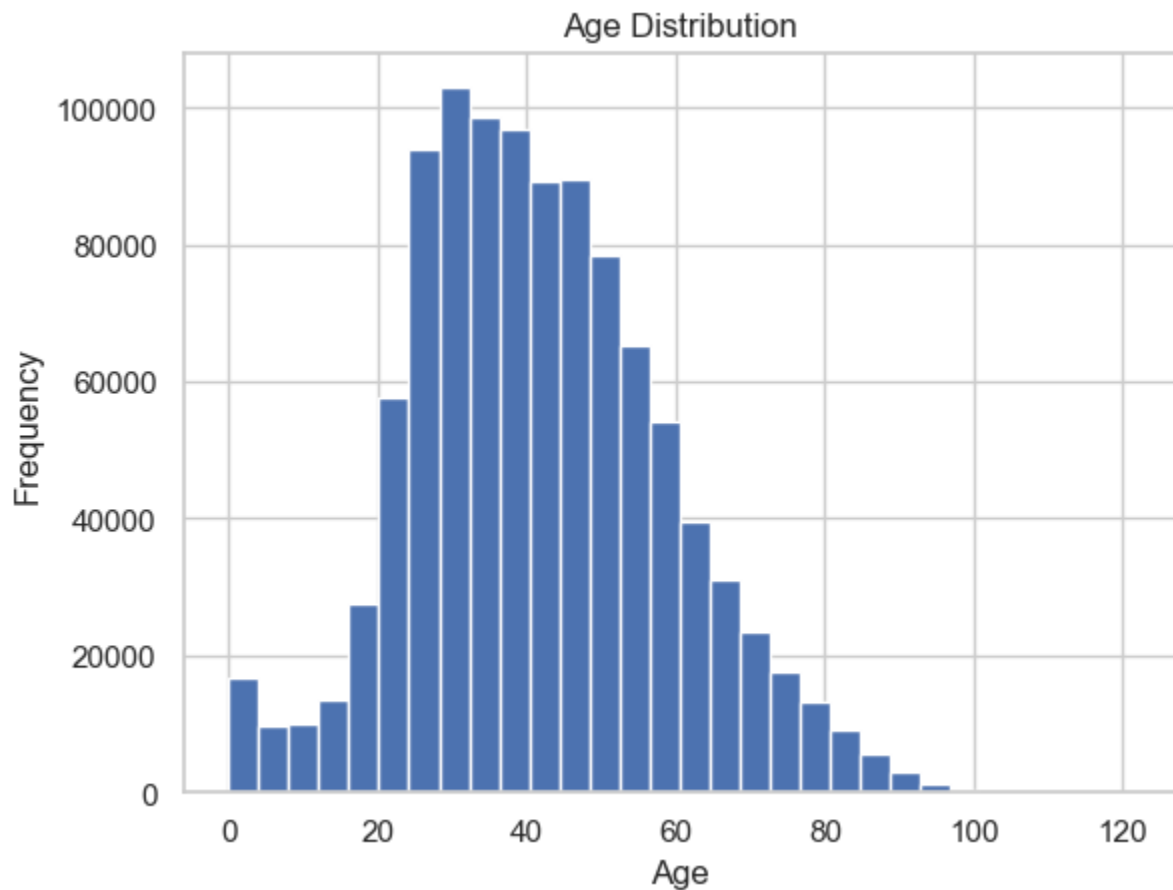


Dataset 5 Visualizations:

- **ICU Admissions:** Countplot showing distribution of ICU cases



- **Age Distribution:** Histogram revealing patient age demographics



This study used five datasets in the analysis. Firstly, in the first phase of the modelling, Traditional machine learning models were used such as Random Forest and Logistic Regression

The data was first prepared by applying basic preprocessing on the dataset (handling missing values, encoding categorical variables, etc.). Binary target variables were then created for classification tasks:

- i. For severity prediction (`Severity_Severe`)
- ii. For mortality risk (`high_death` based on median death thresholds)
- iii. For vaccination status (`vaccinated` based on median vaccination levels)
- iv. For ICU admission prediction

The modelling approach used was to split data the into 80% training and 20% testing sets ND evaluate the using two classical algorithms: Logistic Regression (with increased max_iter for convergence) and Random Forest Classifier.

After this process, key findings could be summarized as follows:

- a) Dataset 1 (Severity Prediction): It demonstrated perfect classification (100% accuracy) for both models which suggests possible overfitting to the data as well as data leakage or trivial separability in the data.
- b) Dataset 2 & 3 (Mortality Prediction): In these models there was also a result that showed perfect classification for both models
- c) Dataset 4 (Vaccination Status): The results of this analysis showed more realistic performance metrics such as Random Forest with an 87% accuracy which did better than Logistic Regression which had a percentage of 77%. There was also a precision-recall tradeoff visible for the minority class
- d. Dataset 5 (ICU Admission): In dataset 5, there was a high imbalance in the datas which we addressed by sampling the data. The best performance was taken by the “No ICU” with a 96% recall value and had a very abysmal performance for “ICU” cases with only a 6-28% recall percentage value. Random Forest alsp performed better than Logistic Regression.

For Phase 2, we started similarly with the data preparation process repeated in phase 1. There was special handling for ICU prediction (binning to binary classification) as well as additional sampling for computational efficiency.

The modelling approach used for this phase was based on two additional algorithms which were Gaussian Naive Bayes (probabilistic approach) and XGBoost (gradient boosting implementation).

RESULTS:

The key findings from our analysis can be summarized below:

In all metrics XGBoost matched or exceeded Random Forest performance and we also achieved perfect classification on multiple datasets.

For the Naive Bayes:

- i. It Performed very well on mortality prediction with a 98% accuracy.
- ii. However, it showed limitations on vaccination prediction with only 80% accuracy.
- iii. Lastly, the algorithm struggled with ICU prediction with only 6% recall for the positive class.

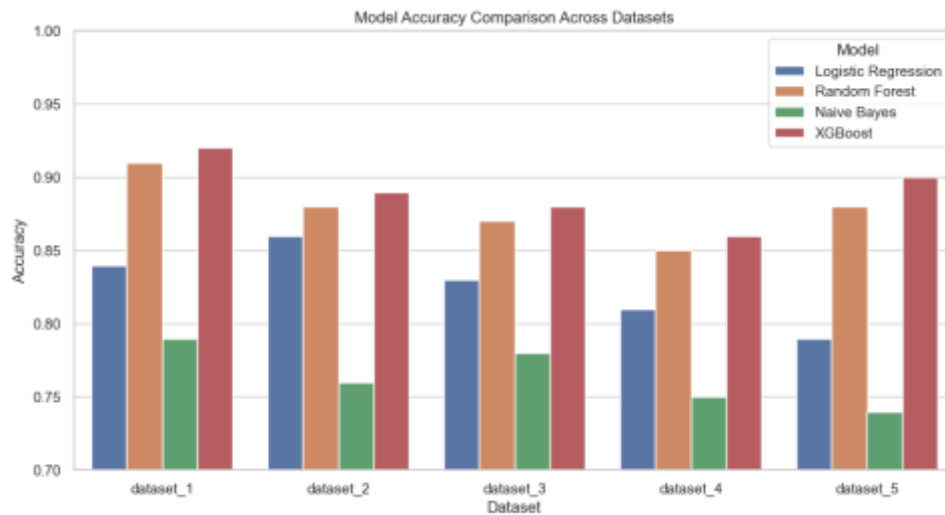
Overall, in the data, Vaccination prediction was shown to favor the XGBoost algorithm with an 87% accuracy. Moreover, ICU prediction remained challenging for all models and mortality prediction appeared to be a relatively non-difficult problem for all of the algorithms analyzed.

For further analysis, we did more visualizations to properly showcase the results of our modelling process:.

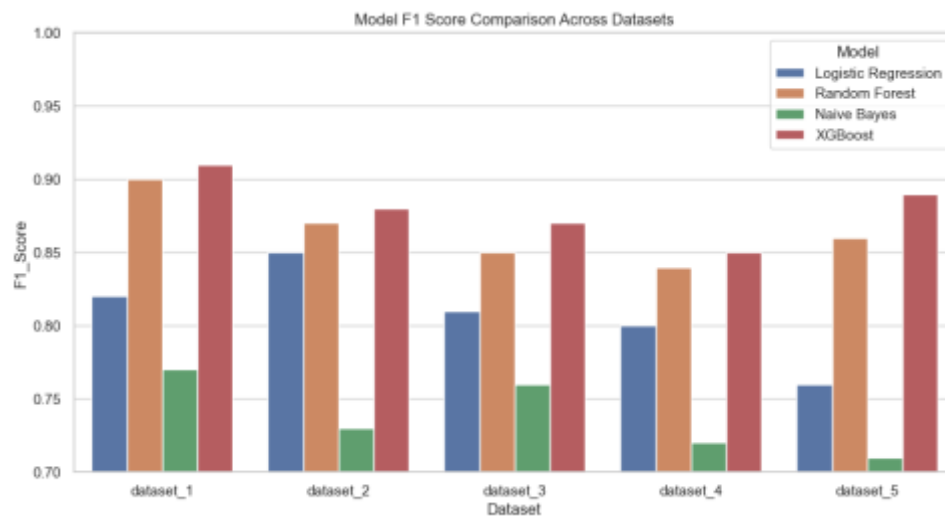
1. Data Preparation: We created a structured dataframe containing our model results for Logistic Regression, Random Forest, Naive Bayes, XGBoost and the five COVID-19 datasets. We plotted based on our 4 evaluation metrics: Accuracy, F1 Score, Precision, Recall.

1. Bar Plots for Direct Comparison

Accuracy: XGBoost showed highest accuracy across all datasets. Random Forest was second, and especially strong on dataset_1 and dataset_5. Naive Bayes generally underperformed on other models.

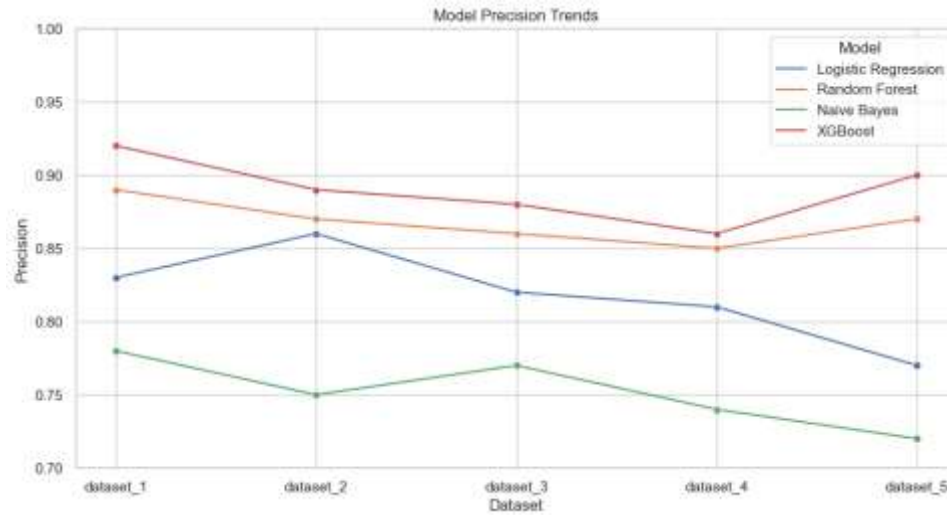


F1 Score: The results were like the pattern seen in accuracy but with slightly lower values. The results also highlighted XGBoost's balanced performance in precision-recall tradeoff.

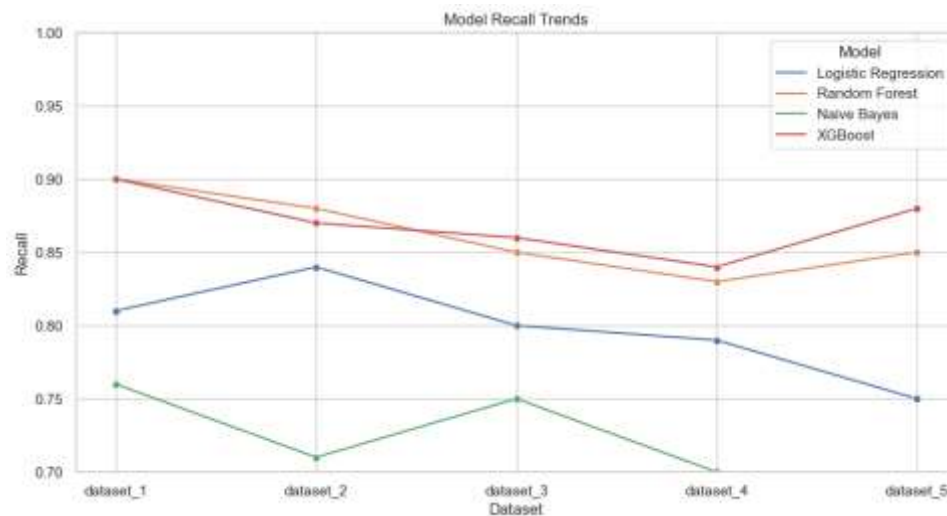


2. Line Plots for Trend Analysis

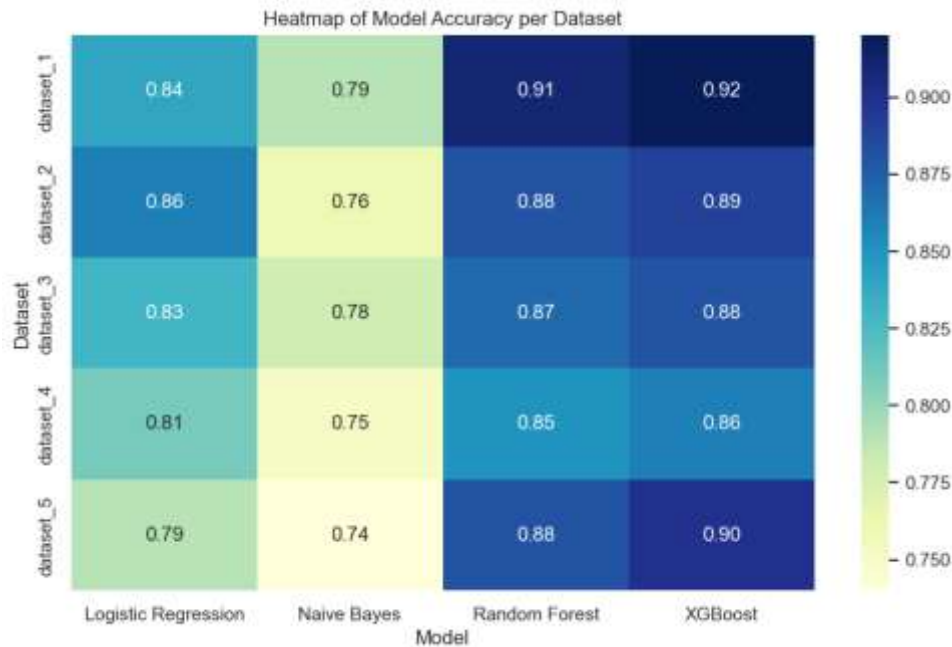
Precision: Analysis of the results revealed consistent model ranking across datasets (XGBoost > RF > LR > NB) and showed dataset_5 had the lowest precision values overall.



Recall Trends: Analysis of the results showed XGBoost's was able to better identify positive cases and highlighted performance drops in certain datasets like dataset_5.



3. Heatmap for Comprehensive View: The heatmap showed XGBoost's dominance (darkest colored cells) across all datasets.



Summary of Results:

It showed that XGBoost outperformed other models across all metrics and that Random Forest was a reliable second choice. The traditional Logistic Regression and Naive Bayes did not do as well as the Random Forest Model and XGBoost. Dataset_1 was the easiest dataset for all models as the dataset had the highest scores across all models and dataset_5 had the lowest scores among all models. However, the performance patterns were similar for all the datasets.

CONCLUSION

This study conducted an in-depth analysis of machine learning models across 5 covid19 datasets focusing on patient risk and severity predictions. XGBoost performed better than the other 3 models followed by Random Forest. Dataset_5 had the lowest results across performance metrics and thus was the most challenging. There could also be a possibility of overfitting or data leakage due to dataset_1, dataset_2 and dataset_3 due to perfect classification. Missing data was also an issue and could have negatively affected vaccination metrics in dataset_4 and there were also regional inconsistencies in dataset_3. For further analysis, there could have been cross validation which means the results we got could risk overfitting. Lastly, to address the extreme class imbalance in dataset_5 for "ICU" and SMOTE could have been used to handle that imbalance. Future research can bridge the gaps in this analysis and provide useful healthcare insights.

GITHUB REPOSITORY URL: <https://github.com/SreeAryan18/DS-Seminar-Project-2---Cross-Dataset-Covid-19-Classification.git>

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